DIGITAL NOTES • SOFT COMPUTING

UNIT – II

Fuzzy Systems

What is inside this unit

1	Fuzzy sets – membership, terms, operations, properties
2	Fuzzy relations – composition, properties
3	Fuzzy logic – linguistic variables, hedges, reasoning
4	Fuzzy rule-based systems – architecture, Mamdani vs Sugeno, defuzzification
5	Solved numerical problems
6	Revision sheet & exam questions

After this unit you will be able to

- Represent concepts with fuzzy sets and apply union, intersection, complement.
- Compute max–min composition of fuzzy relations.
- Use linguistic variables, hedges and approximate reasoning.
- Explain the architecture of a fuzzy system and solve a Mamdani example with defuzzification.

How to use these notes

Read the **blue headings** first, study each diagram, then read the points. Yellow boxes = exam tips / key formulas. Green boxes = quick memory tricks. The last page has a one-page revision sheet and likely exam questions.

1 Fuzzy Sets

A **crisp (classical) set** has sharp boundaries: an element is either **in (1)** or **out (0)**. A **fuzzy set** (Zadeh, 1965) allows **partial membership** – each element has a **membership degree μ between 0 and 1**.

Crisp set	Fuzzy set
μ(x) ∈ {0, 1}	μ(x) ∈ [0, 1]
Sharp boundary	Gradual / soft boundary
"Age ≥ 18 is adult"	"Young", "Old", "Tall" – no exact cut-off
Obeys excluded-middle: A ∪ A′ = X	A ∪ A′ ≠ X in general

1.1 Notation

- Discrete universe: **A = { μ(x₁)/x₁ + μ(x₂)/x₂ + … }** – the "/" and "+" are only symbols, not division/addition.
- Example: **Young = {1/10 + 0.8/20 + 0.5/30 + 0.1/40}** means a 20-year-old is "young" to degree 0.8.
- Continuous universe: A = ∫ μ(x)/x.

1.2 Important terms of a fuzzy set

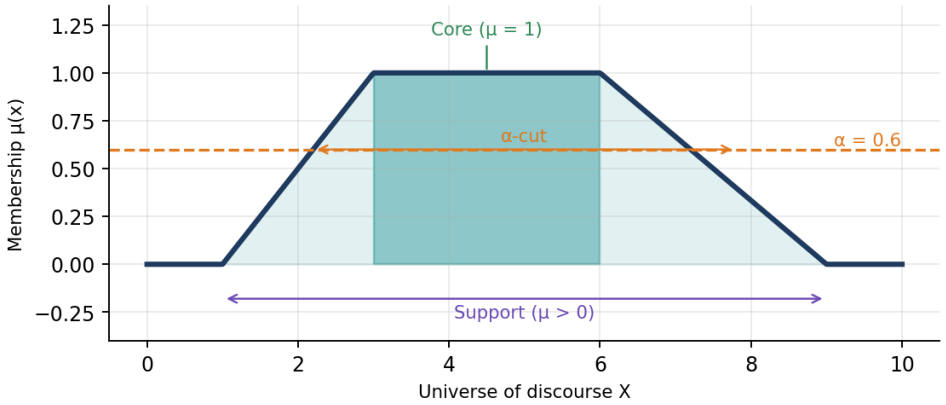


Fig 1.1 Support, core and α-cut of a fuzzy set

Term	Definition	Value in Fig 1.1
Support	All x with μ(x) > 0	[1, 9]
Core	All x with μ(x) = 1	[3, 6]
Height	Maximum membership value	1 (so the set is **normal**)
α-cut (A_α)	Crisp set of x with μ(x) ≥ α	α = 0.6 → [2.2, 7.8]
Crossover point	x where μ(x) = 0.5	—
Cardinality	|A| = Σ μ(x) (scalar)	—

1.3 Membership functions (MF)

The **membership function** maps every element to its degree of membership. Choose the shape by the application; triangular and trapezoidal are the most popular because they are **simple and fast**.

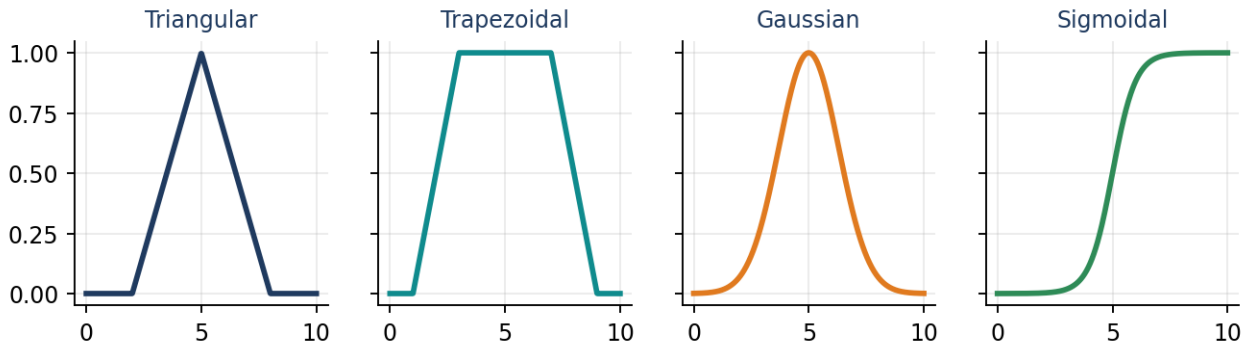


Fig 1.2 Common membership function shapes

- **Triangular** (a, b, c): rises from a to peak b, falls to c.
- **Trapezoidal** (a, b, c, d): flat top between b and c.
- **Gaussian**: smooth bell, $\mu = \exp(-(x-c)^2/2\sigma^2)$ – good for smooth control.
- **Sigmoidal**: S-shaped, used for "open-ended" terms like *very high*.

1.4 Operations on fuzzy sets

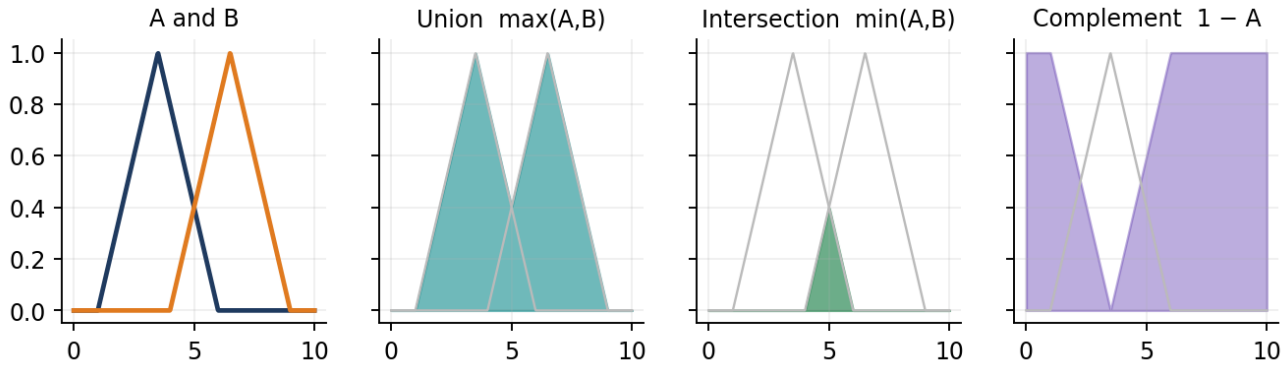


Fig 1.3 Standard fuzzy operations (Zadeh)

Operation	Formula (for every x)	Meaning
Union A ∪ B	$\mu = \max[\ \mu_A(x), \mu_B(x)\ ]$	OR
Intersection A ∩ B	$\mu = \min[\ \mu_A(x), \mu_B(x)\ ]$	AND
Complement A′	$\mu = 1 - \mu_A(x)$	NOT
Algebraic product	$\mu_A(x) \cdot \mu_B(x)$	soft AND
Algebraic sum	$\mu_A + \mu_B - \mu_A\mu_B$	soft OR
Bounded sum	$\min[1, \mu_A + \mu_B]$	—
Bounded difference	$\max[0, \mu_A + \mu_B - 1]$	—

Worked example. A = {0.2/a + 0.7/b + 1/c}, B = {0.5/a + 0.4/b + 0.6/c}

- A ∪ B = {0.5/a + 0.7/b + 1/c}
- A ∩ B = {0.2/a + 0.4/b + 0.6/c}
- A′ = {0.8/a + 0.3/b + 0/c}

Worked example – α-cut and hedges

Tall = {0.1/160 + 0.4/165 + 0.7/170 + 0.9/175 + 1/180}
α-cut (α = 0.5): keep elements with μ ≥ 0.5 → **{170, 175, 180}**
very Tall = μ² → {0.01/160 + 0.16/165 + 0.49/170 + 0.81/175 + 1/180}
somewhat Tall = √μ → {0.32/160 + 0.63/165 + 0.84/170 + 0.95/175 + 1/180}
Notice: "very" makes the set smaller / stricter, "somewhat" makes it larger.

1.5 Properties

Property	Statement
Commutative	A ∪ B = B ∪ A ; A ∩ B = B ∩ A
Associative	(A ∪ B) ∪ C = A ∪ (B ∪ C)
Distributive	A ∪ (B ∩ C) = (A ∪ B) ∩ (A ∪ C)
Idempotent	A ∪ A = A ; A ∩ A = A
De Morgan	(A ∪ B)′ = A′ ∩ B′ ; (A ∩ B)′ = A′ ∪ B′
Involution	(A′)′ = A
Not valid	A ∪ A′ ≠ X and A ∩ A′ ≠ ∅ (excluded-middle & contradiction fail)

Exam tip

The only laws that **fail** for fuzzy sets are **A ∪ A′ = X** and **A ∩ A′ = ∅**. Write this line in every answer on properties.

2 Fuzzy Relations

A **relation** links elements of two sets. A **crisp relation** says "related (1) or not (0)". A **fuzzy relation** R on X × Y gives a **strength of relationship $\mu_R(x, y) \in [0, 1]$**, stored as a **matrix**.

	y_1	y_2	y_3
x_1	0.3	0.7	0.2
x_2	0.8	0.4	0.5

*Fuzzy relation R from $X = \{x_1, x_2\}$ to $Y = \{y_1, y_2, y_3\}$. Example: "x is **close to** y".*

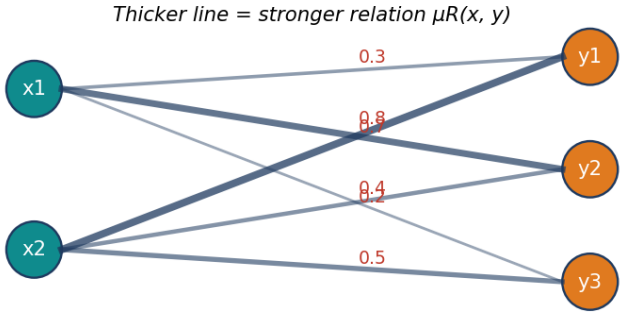


Fig 2.1 Graph view of the fuzzy relation R (weights are membership values)

2.1 Operations on fuzzy relations

- **Union**: R ∪ S = max(R, S) **Intersection**: R ∩ S = min(R, S) **Complement**: R′ = 1 − R
- **Cartesian product** of fuzzy sets A and B: $\mu_{A\times B}(x, y) = \min[\ \mu_A(x), \mu_B(y)\ ]$
- **Inverse** R^{-1}: transpose of matrix R.

2.2 Composition of fuzzy relations

Given R on X × Y and S on Y × Z, the composition **R ∘ S** on X × Z tells how x is related to z **through** y.

- **Max–min:** $\mu_{R\circ S}(x, z) = \max_y \{\ \min[\ \mu_R(x, y), \mu_S(y, z)\ ]\ \}$ (most used)
- **Max–product:** $\mu_{R\circ S}(x, z) = \max_y \{\ \mu_R(x, y) \cdot \mu_S(y, z)\ \}$

Worked example – max–min composition

R (2×3) = [0.30 0.70 0.20] | [0.80 0.40 0.50]
S (3×2) = [0.60 0.90] | [0.20 0.50] | [0.70 0.10]
Step: $(R\circ S)(x_1,z_1) = \max[\ \min(0.3,0.6), \min(0.7,0.2), \min(0.2,0.7)\ ] = \max[0.3, 0.2, 0.2] =$ **0.3**
Repeat for every cell → **R∘S = [0.30 0.50] | [0.60 0.80]** (max–product gives [0.18 0.35] | [0.48 0.72])
*Method: it is like matrix multiplication – replace "×" by **min** and "+" by **max**.*

2.3 Properties of a fuzzy relation (on X × X)

Property	Condition
Reflexive	$\mu_R(x, x) = 1$ for all x
Symmetric	$\mu_R(x, y) = \mu_R(y, x)$
Transitive (max–min)	$\mu_R(x, z) \geq \max_y \min[\ \mu_R(x, y), \mu_R(y, z)\ ]$
Fuzzy equivalence	Reflexive + symmetric + transitive (used for fuzzy clustering)
Fuzzy tolerance	Reflexive + symmetric only

3 Fuzzy Logic

Fuzzy logic is a many-valued logic where a statement can be **partly true**. It lets machines reason with human words like “fairly hot” or “very fast”.

3.1 Linguistic variables

A **linguistic variable** takes **words** as values; each word is a fuzzy set. Example: variable **Temperature** with terms COLD, WARM, HOT.

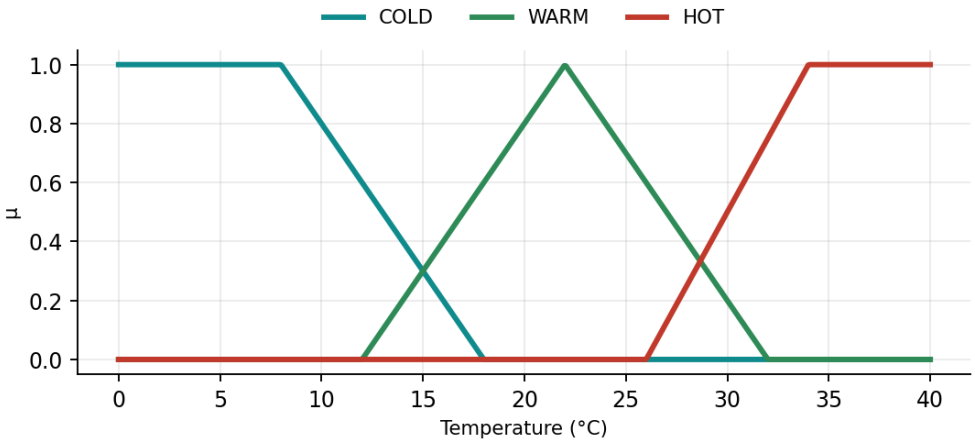


Fig 3.1 Linguistic variable “Temperature” and its three fuzzy terms

3.1b Fuzzy numbers

- A **fuzzy number** is a normal, convex fuzzy set on the real line, e.g. “**about 5**” = triangular (3, 5, 7).
- **Addition** of triangular numbers $(a_1,b_1,c_1) + (a_2,b_2,c_2) = (a_1+a_2, b_1+b_2, c_1+c_2)$. Example: about 5 + about 2 = (3,5,7)+(1,2,3) = **(4, 7, 10)**.
- **Extension principle:** lets any crisp function f(x) be applied to fuzzy sets – $\mu_{f(A)}(y)$ = max over all x with f(x)=y of $\mu_A(x)$.

3.2 Linguistic hedges

Hedge	Operation on μ	Example
very	μ^2 (concentration)	very tall = $(\mu_{tall})^2$
extremely	μ^3	extremely tall
somewhat / more or less	$\sqrt{\mu}$ (dilation)	somewhat tall = $\sqrt{\mu_{tall}}$
not	$1 - \mu$	not tall
slightly	$\sqrt{\mu} \cap$ not(very μ)	slightly tall

3.3 Fuzzy propositions and connectives

- **Atomic:** “x is A” → truth = $\mu_A(x)$. **AND** = min, **OR** = max, **NOT** = $1 - \mu$.

- **Fuzzy implication** "IF p THEN q": **Mamdani** $\min(\mu_p, \mu_q)$; **Larsen** product $\mu_p \cdot \mu_q$; **Zadeh** $\max[\min(\mu_p, \mu_q), 1 - \mu_p]$; **Kleene-Dienes** $\max(1 - \mu_p, \mu_q)$.

3.4 Approximate reasoning – Generalised Modus Ponens (GMP)

	Classical Modus Ponens	Generalised Modus Ponens (fuzzy)
Rule	IF x is A THEN y is B	IF x is A THEN y is B
Fact	x is A	x is **A′** (close to A)
Conclusion	y is B	y is **B′** (close to B)

The conclusion is computed by the **compositional rule of inference**: **B′ = A′ ∘ R**, where R is the fuzzy relation made from the implication A → B.

> **Remember**
> Classical logic needs an exact match of fact and rule. Fuzzy reasoning works even when the fact is only approximately the same – this is why it suits real life.

4 Fuzzy Rule-Based Systems (FRBS)

A **Fuzzy Rule-Based System** (Fuzzy Inference System / Fuzzy Logic Controller) turns **crisp inputs → crisp output** using IF–THEN rules written in words.

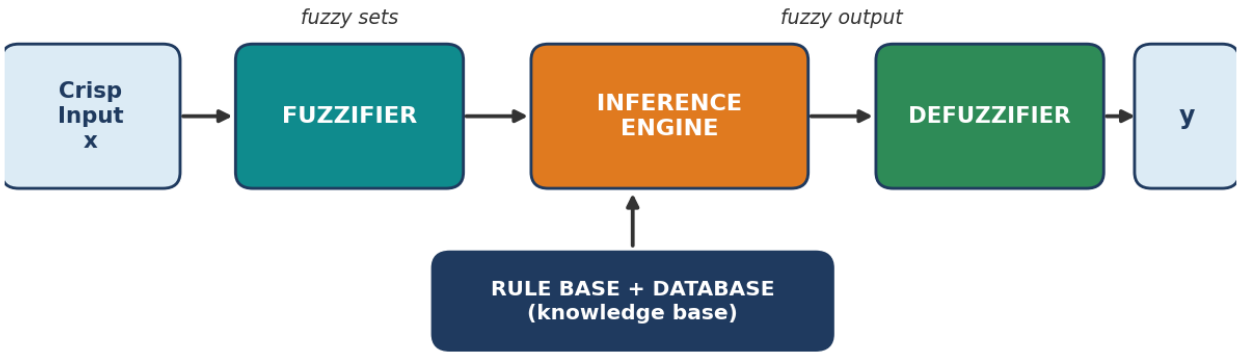


Fig 4.1 Block diagram of a fuzzy rule-based system

Block	Job	Example
Fuzzification	Convert crisp input into membership degrees	28 °C → WARM 0.4, HOT 0.25
Knowledge base	Database (MFs) + rule base (IF–THEN rules from experts)	IF temp is HOT THEN fan is FAST
Inference engine	Fire the rules, combine their results	min for AND, max to combine rules
Defuzzification	Convert fuzzy output into one crisp number	Centroid → one crisp fan speed

4.1 Steps of fuzzy inference

- **Step 1 – Fuzzify** the inputs using membership functions.
- **Step 2 – Apply operators** (AND = min, OR = max) to get each rule's **firing strength**.
- **Step 3 – Implication:** clip (min) or scale (product) each rule's output fuzzy set by its firing strength.
- **Step 4 – Aggregate** all clipped outputs using max.
- **Step 5 – Defuzzify** the aggregated set to a crisp value.

4.2 Mamdani vs Sugeno (TSK) models

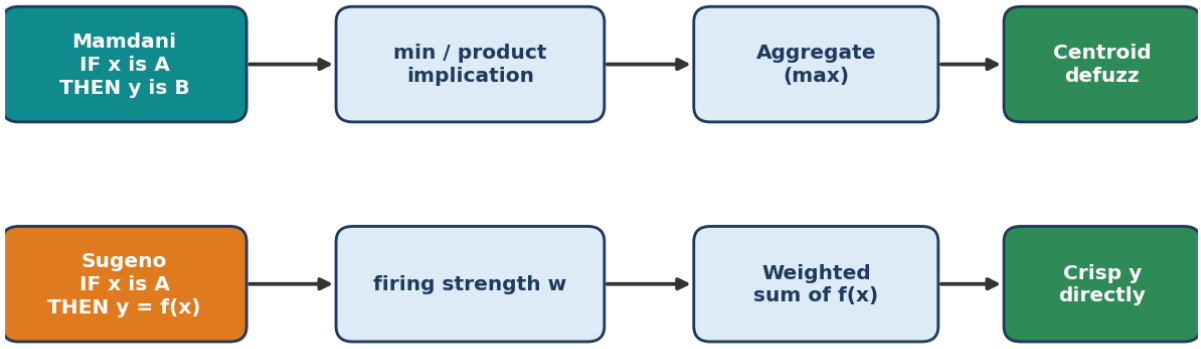


Fig 4.2 Mamdani vs Sugeno inference paths

Point	Mamdani	Sugeno (Takagi–Sugeno–Kang)
Rule consequent	Fuzzy set (IF … THEN y is HIGH)	Function of inputs: $y = a \cdot x_1 + b \cdot x_2 + c$ (or a constant)
Defuzzification	Needed (centroid etc.) – slower	Weighted average $\Sigma w_i y_i / \Sigma w_i$ – fast
Interpretability	High – very human-like	Lower
Best for	Expert-based control, decision support	Adaptive / optimised systems, neuro-fuzzy (ANFIS)

4.3 Worked example – Fan speed controller

Rules: **R1** IF Temp is COLD THEN Fan is SLOW. **R2** IF Temp is WARM THEN Fan is MEDIUM. **R3** IF Temp is HOT THEN Fan is FAST. Input: **28 °C**. Memberships: COLD = 0.00, WARM = 0.40, HOT = 0.25. Each rule clips its output set at its firing strength; the clipped sets are joined with max.

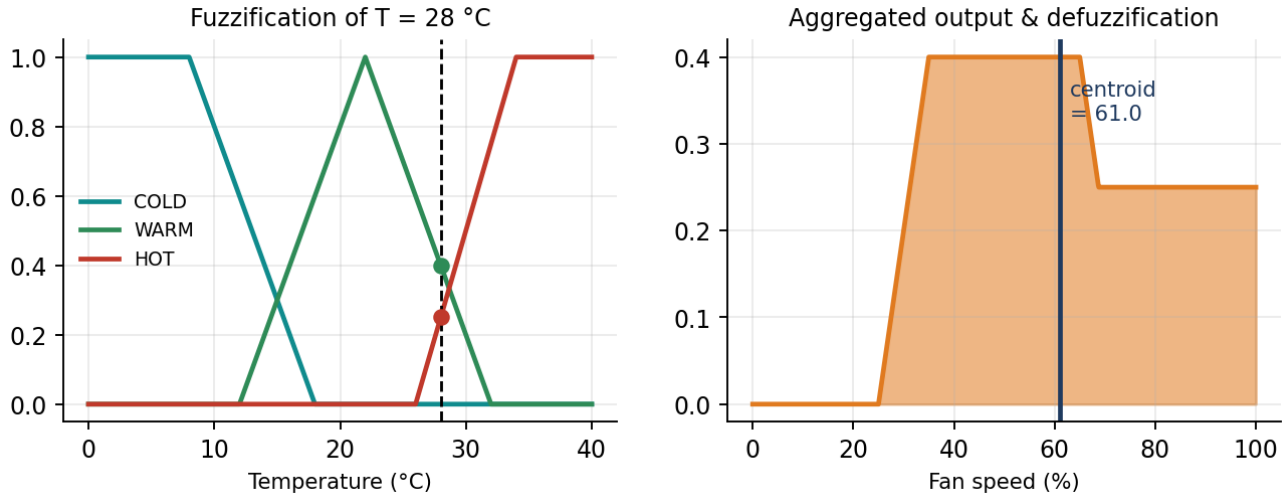


Fig 4.3 Mamdani inference for 28 °C – crisp fan speed ≈ 61 %

4.4 Rule table (FAM) – two-input example

With two inputs the rules are shown as a **Fuzzy Associative Memory (FAM) table**: each cell is one rule, e.g. *IF Service is Good AND Food is Good THEN Tip is MEDIUM*. 3 × 3 terms = 9 rules.

	Food: Poor	Food: Good	Food: Great
Service: Poor	LOW	LOW	MEDIUM
Service: Good	LOW	MEDIUM	HIGH
Service: Great	MEDIUM	HIGH	HIGH

Fig 4.4 FAM table for a tip-advisor fuzzy system

4.5 Defuzzification methods

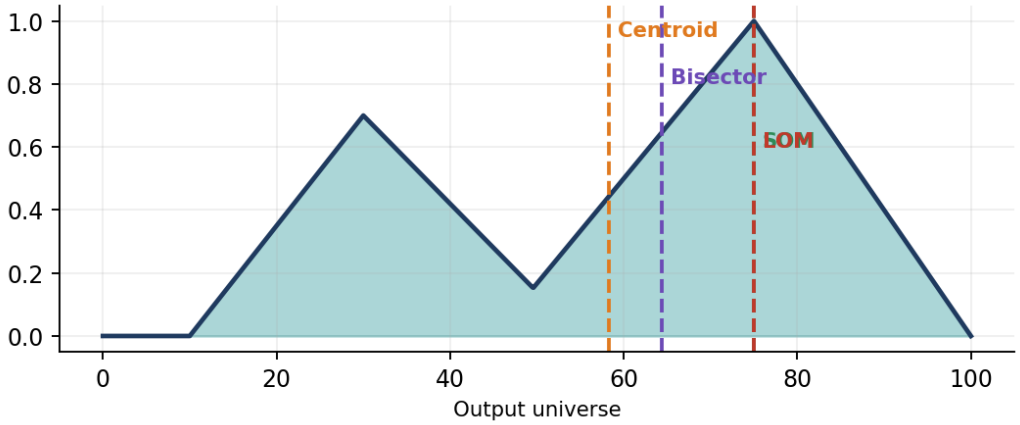


Fig 4.5 Different defuzzifiers give different crisp values for the same fuzzy output

Method	Idea	Formula / note
Centroid (COG, COA)	Centre of gravity of the area – most popular, smooth	$z^* = \int \mu(z) \cdot z \, dz / \int \mu(z) \, dz$
Bisector	Line dividing the area into two equal halves	area left = area right
Mean of Maxima (MOM)	Average of x where μ is maximum	fast, less smooth
Smallest / Largest of Maxima	Left-most / right-most peak point	SOM, LOM
Weighted average	Used with singleton / Sugeno outputs	$z^* = \Sigma \mu_i z_i / \Sigma \mu_i$

Worked example – Sugeno (zero-order) model

R1: IF speed is LOW THEN brake = 10 R2: IF speed is HIGH THEN brake = 80
For one input: firing strengths $w_1 = 0.3$ (LOW), $w_2 = 0.6$ (HIGH)
Output = $(w_1 \cdot 10 + w_2 \cdot 80) / (w_1 + w_2) = (3 + 48) / 0.9 = 56.7$
No clipping, no centroid – just a weighted average, so it is very fast.

4.6 Design steps of a fuzzy controller

- **1.** Identify inputs, outputs and their ranges (universe of discourse).
- **2.** Choose the number and shape of membership functions (3–7 terms is usual).
- **3.** Write the rule base from expert knowledge or data.
- **4.** Select operators: AND (min), implication (min), aggregation (max), defuzzifier (centroid).
- **5.** Simulate, then **tune** MFs and rules (manually or by GA / neural learning).

4.7 Fuzzy controller in a feedback loop

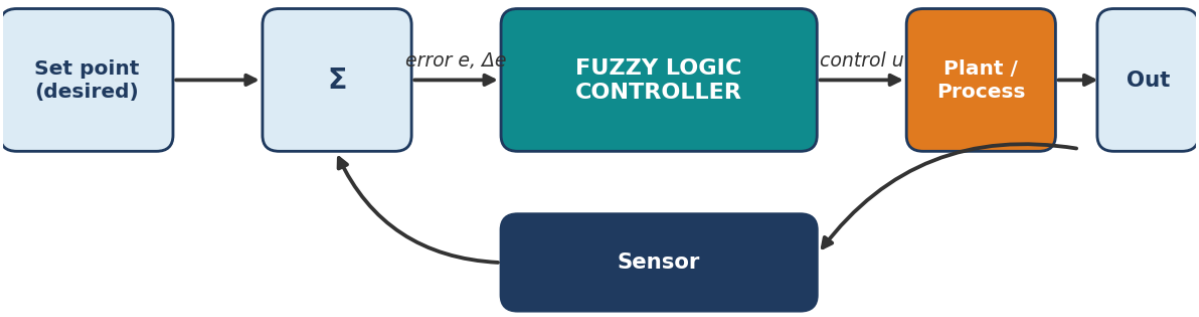


Fig 4.6 Closed-loop control: inputs are usually error (e) and change of error (Δe)

Point	Conventional (PID) controller	Fuzzy logic controller
Needs plant model?	Yes – equations of the process	No – uses expert rules
Non-linear plants	Poor unless re-tuned	Handles well
Noise / sensor errors	Sensitive	Robust
Design effort	Mathematical analysis	Rule writing + tuning
Stability proof	Well established	Difficult

4.8 Advantages, limitations, applications

- **Advantages:** no exact model needed; easy to understand rules; tolerant to noisy sensor data; cheap to implement.
- **Limitations:** rules and MFs are designed by trial/expert; many inputs → **rule explosion** (n inputs × m terms = m^n rules); cannot learn alone.
- **Applications:** washing machine, AC, camera focus, ABS braking, metro control, medical diagnosis, traffic lights.

5 Solved Numerical Problems

Problem 1 – Fuzzy set operations

A = {0.1/1 + 0.6/2 + 1/3 + 0.4/4}, B = {0.5/1 + 0.2/2 + 0.9/3 + 0.7/4}. Find A ∪ B, A ∩ B, A′, and A ∩ B′.
A ∪ B = {0.5/1 + 0.6/2 + 1/3 + 0.7/4} **A ∩ B** = {0.1/1 + 0.2/2 + 0.9/3 + 0.4/4}
A′ = {0.9/1 + 0.4/2 + 0/3 + 0.6/4} **B′** = {0.5/1 + 0.8/2 + 0.1/3 + 0.3/4}
A ∩ B′ = min(A, B′) = {0.1/1 + 0.6/2 + 0.1/3 + 0.3/4}

Problem 2 – Approximate reasoning (compositional rule of inference)

Rule: IF x is A THEN y is B, with A = {0.2/x_1 + 0.8/x_2 + 1/x_3}, B = {0.3/y_1 + 1/y_2}. Fact: x is A′ = {0.1/x_1 + 0.9/x_2 + 0.6/x_3}. Find B′.
Step 1 – relation (Mamdani): R(x,y) = min[A(x), B(y)] → rows x_1,x_2,x_3: [0.2 0.2] | [0.3 0.8] | [0.3 1.0]
Step 2 – compose: B′ = A′ ∘ R (max–min).
B′(y_1) = max[min(0.1,0.2), min(0.9,0.3), min(0.6,0.3)] = **0.3**
B′(y_2) = max[min(0.1,0.2), min(0.9,0.8), min(0.6,1)] = **0.8**
Answer: B′ = {0.3/y_1 + 0.8/y_2}

Problem 3 – Check a relation

R = [[1, 0.8],[0.8, 1]]. Reflexive? diagonal = 1 → **yes**. Symmetric? R_{12} = R_{21} = 0.8 → **yes**. Transitive? R_{12} = 0.8 ≥ max[min(R_{11},R_{12}), min(R_{12},R_{22})] = 0.8 → **yes**. So R is a **fuzzy equivalence relation**.

How to attempt a numerical in the exam

1) Write the formula. 2) Show one full calculation. 3) Give the final answer in the same set notation. Neat steps earn marks even if the final number is wrong.

6 One-page Revision & Important Questions

Formula sheet

• Union = max, Intersection = min, Complement = 1 − μ. • Support: μ>0; Core: μ=1; α-cut: μ≥α.
• Max–min composition: $(R \circ S)(x,z) = \max_y \min[R(x,y), S(y,z)]$. • very = μ^2, somewhat = $\sqrt{\mu}$.
• FRBS: Fuzzify → Fire rules → Aggregate → Defuzzify. • Centroid $z^* = \Sigma\mu(z)\cdot z / \Sigma\mu(z)$.
• Mamdani = fuzzy consequent; Sugeno = linear/constant consequent (weighted average).

Quick self-check

Q1. Which two Boolean laws fail for fuzzy sets? → **Excluded middle and contradiction**
Q2. Which operator replaces AND in fuzzy logic? → **min** (or product)
Q3. Which defuzzifier is most popular? → **Centroid**
Q4. Which model needs no defuzzifier? → **Sugeno**

Likely exam questions

- Define fuzzy set. Differentiate crisp and fuzzy sets. Explain support, core, α-cut with a diagram.
- Explain fuzzy set operations and properties with examples.
- What is a fuzzy relation? Find max–min composition of the given relations (numerical).
- Explain linguistic variables and hedges; state generalised modus ponens.
- Draw and explain the architecture of a fuzzy rule-based system.
- Differentiate Mamdani and Sugeno models; explain defuzzification methods.